

Kuiyi (Cory) Gao

Chapel Hill, NC

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kuiyigao.github.io

GitHub

Languages

Mandarin (native), English (TOEFL iBT 110), Cantonese (intermediate).

Research Focus

AI safety, agent reliability, multimodal safety evaluation, runtime monitoring, and human oversight.

Honors and Awards

Professor Joseph Lau Scholarship of Innovation

Faculty of Engineering ELITE Stream Award

CUHK College Entrance Scholarship

CUHK Exchange Scholarship, UNC Spring 2026

Education

University of North Carolina at Chapel Hill

Jan 2026 to present

Undergraduate, intended B.A. in Computer Science

Expected May 2027

GPA 3.68/4.00. Coursework: algorithms, machine learning, NLP, technology ethics, C/Unix systems, and object-oriented design.

The Chinese University of Hong Kong

Sep 2023 to May 2026

Undergraduate study in Computer Science, transferred to UNC

Coursework: graduate deep learning, artificial intelligence, data structures, formal languages, computer organization, probability, statistics, and linear algebra.

Publications and Manuscripts

[1] **Kuiyi Gao*** and collaborators. *RealGUINoise: A Dynamic Cross-Platform Benchmark for GUI Agent Robustness under Real-World Interface Noise*. Co-first author. Under review, 2026.

[2] **Kuiyi Gao**. *RuntimeTickets: Enforcing and Auditing Task-Specific Security Requirements for Third-Party Agent Skills*. First author. Under review at ICLR 2027.

[3] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. ACL 2025 Main Conference. [paper] [code]

[4] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. Findings of ACL 2025. [paper] [code]

Research Experience

RealGUINoise [1]

Hong Kong and remote

CUHK, advisor: Prof. Michael R. Lyu

Sep 2025 to present

Co-first author. Built paired web and Android tasks with controlled interface noise; designed the evaluation protocol and automated judge; ran clean-versus-altered experiments measuring task completion, unsafe actions, and action-sequence changes.

RuntimeTickets [2] [code]

Abu Dhabi and Beijing

MBZUAI UGRIP and continuing research, advisor: Dr. Steve Liu

Jun 2026 to present

First author. Designed task-specific contracts that narrow host permissions, gate runtime requests, and preserve decision and effect records. Built the runtime firewall in a four-student UGRIP team and lead method design, experiments, and writing.

Chain-of-Jailbreak [4]

Hong Kong

Tencent AI Lab and CUHK

Jun 2024 to Feb 2025

Second author. Designed the multi-turn image-editing attack and prompting defense, contributed benchmark data, and ran most attack and defense experiments. Built evaluation scripts and prepared figures, manuscript text, and rebuttal.

MMSafeAware [3]

Hong Kong and Shenzhen

Tencent AI Lab

Dec 2023 to Oct 2024

Third author. Built benchmark data and evaluation for 1,500 image-text pairs across 29 safety scenarios. Ran vision-centric fine-tuning, analyzed missed risks and benign false alarms, and contributed to submission and rebuttal.

Selected Projects

MemSlot: Long-Document Compression [code]

2026

Natural language processing course project

Tests whether learned word saliency can compress contracts before OCR and question answering. Implemented memory-slot attention over frozen RoBERTa features, overlap-aware scoring, saliency-guided rendering, and pruning baselines; end-to-end OCR-to-QA evaluation remains incomplete.

Poetry and Image Variation

2025

AI and language course project

Examines whether poem style and prompt length relate to variation among generated images. Compared saved CLIP-based scores for Imagist and Romantic excerpts generated with FLUX and Stable Diffusion using SciPy and Matplotlib.

Text-KEN: Kernel Comparisons for Sentence Embeddings

May to Aug 2025

CUHK summer research, advisor: Prof. Farzan Farnia

Tests a nonparametric comparison of two text collections without training a classifier. Built a MiniLM embedding pipeline with RBF kernels, spectral entropy, centering, and covariance whitening.

Movability Segmentation [code]

2025

Deep learning course project

Maps scene categories to four movability classes for interpretable semantic segmentation. Implemented DeepLabV3 with a custom ASPP head, COCO-Stuff remapping, image-mask preprocessing, and checkpoint inference.

Technical Skills

Research engineering Python, Git, Linux, Docker, Playwright, HTTP/JSON APIs, experiment configuration, trace logging.

Machine learning PyTorch, Hugging Face Transformers, sentence-transformers, attention, fine-tuning, semantic segmentation.

Evaluation and analysis Benchmark design, task and action metrics, multimodal judgment, span-QA EM/F1, kernel methods, statistical testing.